

KHOUN KHAN, Thailand

WMO: 485700

Lat: 6.6536N

Lon: 100.0831E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	22.8	23.3	18.4	13.2	29.3	19.2	13.9	28.9	4.7	29.6	4.1	30.2	0.0	N/A	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.6	35.5	25.5	34.7	25.6	34.0	25.6	27.4	32.3	27.1	31.9	26.9	31.5	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.6	25.9	21.3	29.3	25.7	20.9	29.1	87.0	32.3	85.6	32.0	84.5	31.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	3.8	3.3	DB	21.0	36.4	1.0	1.0	20.3	37.1	19.7	37.7	19.1	38.3	18.4	39.1	(4)
(5)			WB	19.5	28.3	1.1	0.5	18.7	28.6	18.0	28.9	17.4	29.1	16.5	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	28.3	28.9	29.3	29.4	28.9	28.5	28.1	28.0	27.8	27.8	28.0	28.0	(6)	
	DBStd	1.13	1.01	0.96	1.09	0.97	0.92	0.91	0.91	1.02	1.08	0.96	1.00	1.00	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6724	567	529	599	582	587	556	561	559	534	552	539	558	(10)	
	CDD18.3	3682	309	296	341	332	328	306	303	301	284	293	289	300	(11)	
	CDH23.3	39314	3292	3350	3948	3804	3704	3381	3204	3109	2801	2791	2851	3078	(12)	
(13)	CDH26.7	15389	1275	1518	1876	1700	1503	1332	1196	1124	946	893	963	1063	(13)	
(14)	Wind	WSAvg	0.6	1.1	1.1	0.6	0.4	0.3	0.4	0.5	0.5	0.4	0.5	0.9	(14)	
(15) Precipitation	PrecAvg	2048	51	46	131	169	180	165	190	225	260	287	213	133	(15)	
	PrecMax	2412	295	168	404	304	332	341	333	339	431	511	471	488	(16)	
	PrecMin	1550	7	1	6	24	86	29	75	40	149	132	103	0	(17)	
	PrecStd	202	62	45	93	72	56	65	67	77	71	79	89	98	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.6	36.7	36.5	35.2	34.0	33.5	33.5	33.4	33.3	33.5	33.5	(19)	
		MCWB	24.8	24.4	25.2	26.1	26.8	26.1	26.0	26.0	26.1	25.9	26.1	25.5	(20)	
	2%	DB	33.3	34.6	35.7	35.1	34.0	33.4	32.8	32.7	32.5	32.5	32.7	32.8	(21)	
		MCWB	24.7	24.6	25.2	26.3	26.5	26.1	25.9	25.9	25.9	25.9	25.9	25.3	(22)	
	5%	DB	32.6	33.9	34.9	34.2	33.2	32.7	32.2	32.1	31.9	31.7	32.0	32.1	(23)	
		MCWB	24.6	24.7	25.3	26.4	26.5	26.2	26.0	25.8	25.9	25.9	25.9	25.2	(24)	
	10%	DB	31.9	33.1	33.9	33.3	32.5	32.0	31.5	31.5	31.1	30.9	31.2	31.4	(25)	
		MCWB	24.5	24.5	25.3	26.4	26.4	26.2	26.0	25.8	25.8	25.8	25.7	25.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.6	27.2	28.0	27.7	27.5	27.2	27.1	27.1	27.1	27.0	26.8	(27)	
		MCDB	31.2	32.6	33.4	33.6	32.6	31.9	31.3	31.5	31.3	31.2	31.5	31.4	(28)	
	2%	WB	26.0	26.0	26.7	27.4	27.3	27.1	26.7	26.7	26.7	26.6	26.6	26.2	(29)	
		MCDB	31.0	32.0	32.6	32.8	32.0	31.4	30.8	30.9	30.7	30.6	31.0	31.0	(30)	
	5%	WB	25.5	25.6	26.3	27.1	27.0	26.8	26.5	26.4	26.3	26.2	26.2	25.9	(31)	
		MCDB	30.5	31.5	32.2	32.3	31.7	30.9	30.6	30.5	30.2	30.2	30.5	30.5	(32)	
	10%	WB	25.1	25.1	25.9	26.7	26.7	26.5	26.2	26.1	26.1	26.0	26.0	25.5	(33)	
		MCDB	30.1	31.0	31.7	31.6	31.0	30.5	30.2	30.1	29.9	29.8	30.1	29.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	9.2	9.6	8.4	7.3	7.1	6.9	6.7	6.5	6.5	6.7	6.7	(35)	
		MCDBR	8.6	10.2	11.0	9.5	8.3	8.0	7.9	7.7	7.7	7.7	7.9	8.0	(36)	
	5% WB	MCWBR	2.8	3.0	3.1	2.8	2.6	2.5	2.6	2.4	2.4	2.5	2.6	2.7	(37)	
		MCWBR	7.6	9.2	9.5	8.5	7.6	7.3	7.2	7.1	7.0	6.9	7.3	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.453	0.466	0.476	0.473	0.455	0.463	0.467	0.468	0.458	0.454	0.450	0.454	(40)	
		taud	2.323	2.257	2.257	2.275	2.351	2.349	2.325	2.310	2.335	2.349	2.357	2.336	(41)	
	Ebn at Noon	Ebn at Noon	865	863	854	843	840	821	822	836	856	863	861	856	(42)	
		Edh at Noon	130	143	144	139	125	124	128	132	131	128	125	127	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.08	5.96	5.95	5.77	5.10	4.93	4.83	4.92	4.78	4.65	4.43	4.36	(44)	
	RadStd	RadStd	0.36	0.35	0.51	0.28	0.29	0.25	0.23	0.19	0.19	0.34	0.32	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.22	+0.14	N/A	+0.32	+0.22	+0.19	N/A	N/A	+94	+94

Nomenclature: See separate page